

PAPER 1A

1-

Compounds in mixture interact with solvent as well as chromatographic paper.

Chromatographic paper is polar, solvent should be apolar.

Polar compounds in mixture will interact with chromatographic paper, therefore will be delayed.

2-

The statement above mentions only physical properties such as density and melting point, thus only physical quantities can be commented.

3-

Ionization energy is the energy required to remove only one electron from element at its gaseous phase.

It is easier to remove valance elctrons than core electrons because outermost shell electrons held loosely through Coulombic force.

2744.78 → 11 577.5 5 times greater , biggest difference
element has three valance electrons → third group on periodic table



4-

$$n = m / MM$$

$$n_{\text{Ar}} = \frac{30 \text{ g}}{39,948 \text{ g/mol}} = 7.509762691 \times 10^{-1} \text{ mol}$$

$$n_{\text{Ne}} = \frac{60 \text{ g}}{20.180 \text{ g/mol}} = 2.973240833 \text{ mol}$$

$$\frac{n_{\text{Ar}}}{n_{\text{Ne}}} = \frac{0.7509}{2.973} = 0.2525783518$$

5-

$$PV = nRT \quad P = \frac{nRT}{V}$$

$$\frac{P_f}{P_i} = \frac{N_f * T_f}{N_i * T_i}$$

$$P_1 = \frac{N_f * T_f}{N_i * T_i} * P_i$$

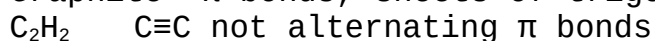
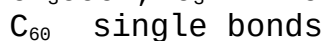
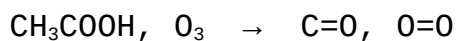
$$P_f = \frac{N/4}{N} * \frac{450 \text{ K}}{300 \text{ K}} * 40 \text{ kPa} = 15 \text{ kPa}$$

6-



7-

Chemical species with delocalized electrons: alternating π bonds, conjugated systems, aromatic ring
double bond with electronegative atoms, electronegative atoms will withdraw electrons toward itself



8-

S central atom, 6 electron groups around central atom
4 groups at equatorial plane, 2 groups at polar sites

9-

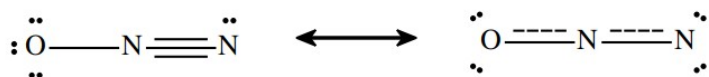
N₂O

Least electronegative atom at the center

$5 \times 2 + 6 = 16$ valance e⁻

Allocate electrons starting from the most EN atom

Complete octets



Resonance structure

The most stable resonance structure will dominate

10-

O has 2 lone pairs of electrons, N has 1 lone pair of electron
Aromatic ring has 6 π electrons, double bond has 2 π electrons

11-

Malleability mould the metal easily
Majority of features of the transition metals arise from free electrons, electron sea model

12-

There must be a π bond between CH₂ and C(CH₃)₂ so that π bond breaks and makes bond with other monomers

13-

Emitted light arrives to our eyes therefore we can see CuSO₄ as blue. Emission occurs when an electron drops from higher energy level to lower energy level.

14-

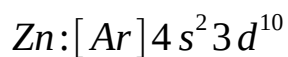
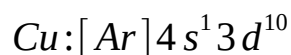
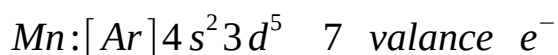
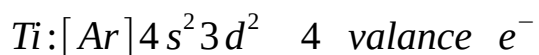
CO is carbonyl group , C=O

15-

What are the names of block?

s block elements , p block of elements , d, f
Orbitals electrons occupy

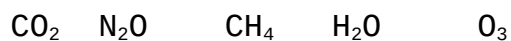
16-



Even if the highest oxidation state numbers of Cu and Zn are not known necessarily, it is known that later transition metal elements have lower oxidation state number.

Thus Cu and Zn can not have oxidation state number which is greater than 7.

17-



CO₂ by burning fossil fuels

All are colourless, can not absorb UV radiation

H₂O has polar bond

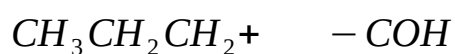
18-

Cyclohexane ring flips rapidly at room temperature, all H₂s become identical

19-



$$48+16+8=72\text{ g/mol}$$



$$43\text{ m/z} \quad 29\text{ m/z}$$

More stable fragments form

Tertiary carbocation can stabilize plus charge through electron donating methyl groups

Carbonyl group can stabilize minus charge through electron withdrawing and delocalization

20-

Energy is released, then it is exothermic reaction

heat is emitted to surroundings, temperature will increase

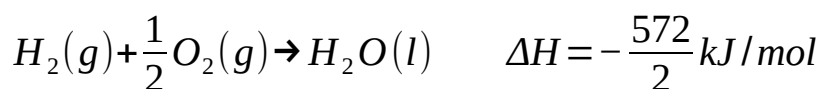
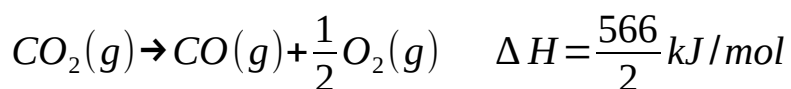
21-

Hess law

If reactions are summed, their enthalpy values are summed

If reverse a reaction, enthalpy value of that reaction changes its sign

If a reaction multiplied by a number, then enthalp value is multiplied by the same number

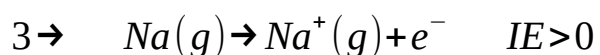
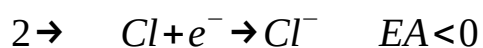
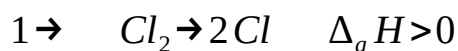


$$\frac{566}{2} - \frac{572}{2} + 44 = 41 kJ/mol$$

23-

Born-Haber cycle is procedure of several consecutive reactions to find lattice energy of a crystal

NaCl



24-

K_c : equilibrium coefficient ratio of concentration at equilibrium, fixed number at given temperature

Q: reaction quotient, ratio of concentration at any time as reaction progresses, it changes as reaction proceeds

At equilibrium $Q = K_c$

ΔG° : standard free energy is free energy of a reaction

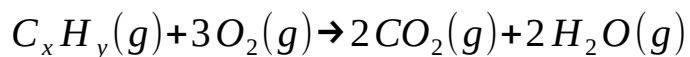
ΔG : Free energy change at any time as reaction proceeds

At equilibrium $\Delta G = 0$

$$\Delta G = \Delta G^\circ + RT \ln Q$$

25-

$$PV = nRT \quad V \sim n$$



20

40

40

$$x=2 \quad y=4$$

26-

Activation energy is minimum energy required to initiate a reaction

$$E_A(\text{forward}) = E(\text{TS}) - E(\text{reactants}) = Y - X$$

$$E_A(\text{reverse}) = E(\text{TS}) - E(\text{products}) = Y - Z$$

27-

Temperature will increase their kinetic energy, so particles will collide more frequently

Chemical reaction occurs when a bond forms between atoms, particular atoms must collide, not every collision results a reaction

28-

Quotient of the concentration of products divided by that of reactants

Stoichiometric coefficient becomes powers of concentration

29-

Summit on reaction profile are TS
pits are intermediate

30-

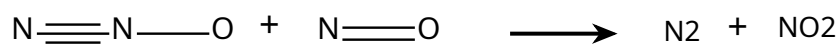
$$K_c=\frac{[C]^2}{[A][B]^2}$$

$$K_c=\frac{2^2}{2*1^2}=2$$

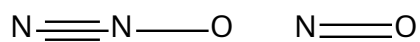
31.

$$k=A*e^{-\frac{E_A}{RT}}$$

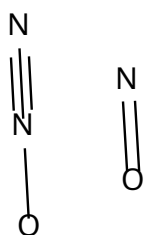
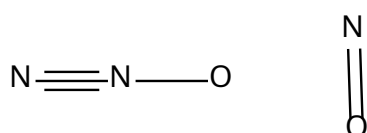
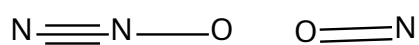
Steric factor



This orientation yields a reaction

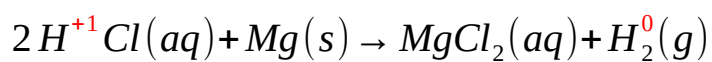


Not proper orientation to give reaction



32.

H transfer → acid base reaction



H is oxidized not proton transfer but oxidation

33-

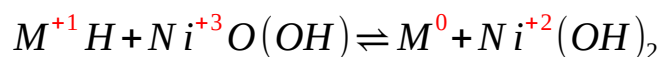
An indicator changes color depending on pH

In acidic solution, $[\text{H}^+]$ will increase, reaction will go in reverse direction → Color A

In basic solution, base will consume $[H^+]$, proton concentration will decrease,
 reaction will go forward $\rightarrow$ color B

If color changes slightly at end point, then titration error will be smaller.

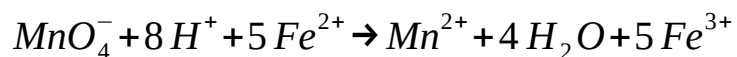
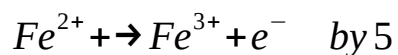
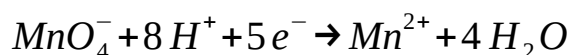
34-



H is both reduced and oxidized.

35-

Separate reduction and oxidation half reactions
 Balance #atoms except O and H
 Add water to offset #O, then add H^+ to offset #H
 Add e^- to balance charge
 Add two half reactions



$$5(M*V)_{Mn} = (M*V)_{Fe}$$

$$5*0.1 M * V = 0.1 M * 20.0 mL$$

$$V = 4.00 mL$$

$$dm^3 = L \quad cm^3 = mL$$

do not convert from ml to L, it will cancel out at both sides

36-

If C-Cl can be broken but not C-F, then C-Cl should be stronger than C-F

If much O_3 are broken, then many radicals must have been produced.

37-

NH_3 can donate lone pair electrons so it is Lewis base

$NH_3 + H^+ \rightarrow NH_4^+$ it can accept proton Bronsted-Lowry base however

$NH_3 + B^- \rightarrow NH_2^- + HB$ if it meets an stronger base than itself then, NH_3 acts as Bronsted-Lowry acid.

It has lone pair electrons it is a nucleophile

It can donate its lone pair electrons to metal cation, It can form coordination covalent bond it is a ligand

38-

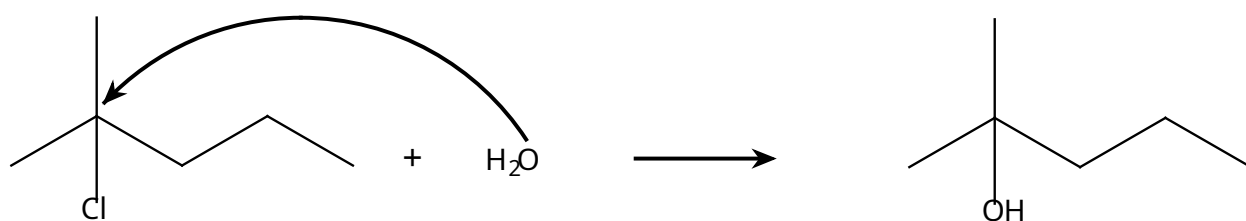
The better LG stabilizes the negative charge better

Hallides

Polarizability

increasing size $\rightarrow$ $F < Cl < Br < I$

39-



Substitution reaction

SN1 tertiary α C

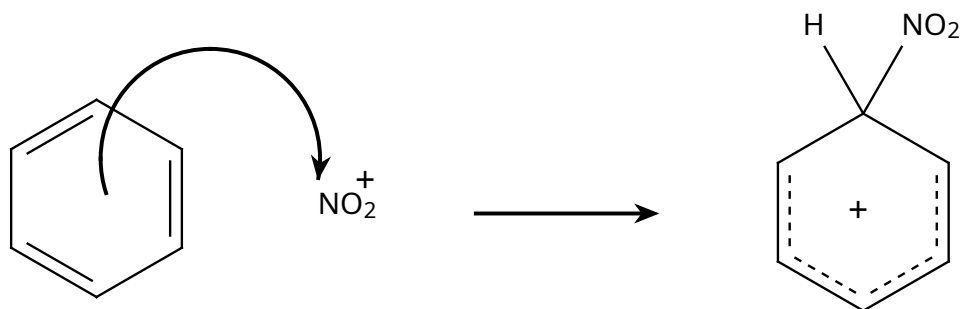
Water attacks to tertiary α C

- 1- Tertiary carbocation will form
- 2- Water will attack to carbocation
- 3- Another water molecule will grab one of the protons

Homolytic bond cleavage -> Radicals

Heterolytic bond cleavage -> cation and anion form

40-



Solid lines mean bond

Dashed lines mean electron delocalization

NO₂⁺ is electrophile

Arenium ion is intermediate

PAPER 1B

1-

a-



O is more EN than Cl so O, O will withdraw electron density towards itself, being partially negative, Cl being partially positive

b-

An equilibrium reaction. Le Chatelier principle states that equilibrium reaction counteracts to an action on reaction

c-

i-

Units are inconsistency.

Measurement conditions might be different.

Significant figures are uncertain

ii-

mass/mass, volume/volume, mass/volume

that of solute / that of solution

$$\frac{177 \text{ cm}^3}{100 \text{ cm}^3}$$

$$M = \frac{n}{V_{\text{solution}}}$$

$$n = \frac{m}{MM}$$

$$d = \frac{m}{V} \quad m = d * V$$

$$n = \frac{d * V_{\text{solute}}}{MM}$$

$$M = \frac{\frac{d * V_{\text{solute}}}{MM}}{V_{\text{solution}}} = \frac{2.86 \frac{\text{g}}{\text{L}} * 0.177 \text{ L}}{70.90 \frac{\text{g}}{\text{mol}} * 0.100 \text{ L}}$$

$$M = 7.139915374 \times 10^{-2} M \approx 0.0714 M$$

iii-

As the temperature increases, solubility of Cl_2 decreases.

Exothermic reaction

Cl_2 is apolar, H_2O polar, intermolecular interactions, i.e. induced dipole-dipole is rather weak.

Thermal energy surmounts intermolecular forces at high temperatures

d-

Elemental Cl_2 is gas. Gaseous matter has low solubility in water, therefore if directly added, Cl_2 gas will not dissolve in water effectively.

Cl_2 might be toxic

e-

i-

Within pH range 0-6

ii-

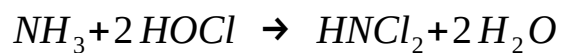
$$K_a = \frac{[H_3O^+][OCl^-]}{[HOCl]}$$

$$pK_a = pH - \log \frac{[OCl^-]}{[HOCl]}$$

$$[OCl^-] = [HOCl] \quad pK_a = pH$$

f-

i-



ii-

Decrease NCl_3

pH range of 5 - 8

HOCl reacts with NH_3 from sweat and urine.

We may consider to decrease concentration of HOCl but it sterilizes swimming pool, hygiene can not be relinquished. It can be preferred a disinfectant that does not produce HOCl

Activation energy is required for any reaction to occur.

At low temperature activation energy threshold may not be surpassed

Reaction between HOCl and NH_3 does not occur

g-

Solubility of Cl_2 at low temperature will be high, Cl_2 dissolves in water, will not be released to environment.

Less energy consumption

Prevent undesired reactions which do not happen at low temperature

2-

a-

Controlled variables are constant factors in each sample

Amount/volume of orange juice

Temperature

Brand of orange juice

b-

i-

To use less titrant, I_2

Orange juice is colorful, if concentrated it will be harder to notice end point, color change of indicator in colorful and concentrated solution

ii-

Some air remained in the flask before it is plugged with stopper. Therefore orange juice is exposed to little amount of air.

Stopper must be opened before titration

iii-

How to remove remained air in the flask?

Pump N_2 or any inert gas to replace air

Seal the opening of the flask with rubber piece, pierce with needle of syringe, withdraw sample, cover the hole with sticky tape

c-

Addition/subtraction add uncertainties

percent uncertainty = uncertainty / value * 100

$$0.05\text{ cm}^3 + 0.05\text{ cm}^3 = 0.1\text{ cm}^3$$

$$16.10\text{ cm}^3 - 1.10\text{ cm}^3 = 15.00\text{ cm}^3$$

$$\% \text{uncertainty} = \frac{0.1\text{ cm}^3}{15.00\text{ cm}^3} * 100 = 60\%$$

d-

i-

Rate = concentration change / time

$$\text{rate} = \frac{1.1 * 10^{-3} - 5.3 * 10^{-3}}{14 \text{ days}} = 0.03 * 10^{-4} \frac{M}{d}$$

ii-

Data supports that because concentration of ascorbic acid 24h sample is lower than 0h sample, as exposure time increases concentration of ascorbic acid decreases.

Data does not support that O₂ destroys ascorbic acid by oxidizing it

iii-

how to avoid loss of vitamin C in orange juice?

Orange juice is exposed to air within different time interval

Close the lid of orange juice container to prevent loss of vitamin C

Consume orange juice in short time

e-

What else could it be inquired?

Temperature, at room temperature or refrigerator

Light, exposed to day light, dark in the refrigerator

Fresh or processed

3-

a-

Initial rate, slope of curve

Average rate, difference between initial and final states

$$\frac{40 \text{ cm}^3}{35 \text{ s}} = 1.14285 \frac{\text{cm}^3}{\text{s}}$$

b-

30s

c-

When pressure increases, volume decreases.

Solubility of gases depends on temperature and pressure

when pressure increases, solubility of CO_2 will increase, therefore volume of CO_2 decreases

d-

$$\frac{V_f - V_i}{V_i} * 100$$

$$33 \text{ cm}^3 - 11.5 \text{ cm}^3 \text{ per } 400 \text{ g}$$

$$\frac{(33 - 11.5)}{4} = 5.375 \text{ cm}^3 \text{ volume difference per } 100 \text{ g}$$

$$\frac{5.375 \text{ cm}^3}{33.0 \text{ cm}^3} * 100 = 16.28 \%$$

e-

Mass of apparatus does not change as time goes.

PAPER 2

1-

a-

Empirical formula vs Molecular formula

100 g HX

39.99 g C 6.73 g H 53.28 g O

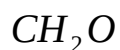
$$n = \frac{m}{MM}$$

$$n_C = \frac{39.99 \text{ g}}{12.000} = 3.3325 \text{ mol}$$

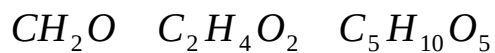
$$n_H = \frac{6.73 \text{ g}}{1.000} = 6.73 \text{ mol}$$

$$n_O = \frac{53.28 \text{ g}}{16.000} = 3.33 \text{ mol}$$

$$n_C \approx 1 \quad n_H \approx 2 \quad n_O \approx 1$$



To find molecular formula molar mass is needed



Compounds above have different molecular formula and the same empirical formula

b-

$$n_a = (MV)_{base}$$

$$n = \frac{m}{MM} \quad MM = \frac{m}{n}$$

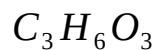
$$MM = \frac{m}{M_b * V_b} = \frac{1.51 \text{ g}}{0.750 \text{ M} * 0.02230 \text{ L}} = 90.28400 \frac{\text{g}}{\text{mol}} \approx 90.3 \frac{\text{g}}{\text{mol}}$$

c-

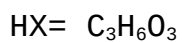
$$CH_2O \quad MM = 90.3 \frac{\text{g}}{\text{mol}}$$

$$12.000 + 2.000 + 16.000 = 30.000 \text{ g}$$

$$\frac{90.3}{30.000} \approx 3$$



d-



Carboxylic group -COOH

e-

i-

Around 1780 cm^{-1} is carbonyl , Z

Around $3000\text{-}3500\text{cm}^{-1}$ is hydrogen bonding, Y

ii-

